EEEE4138 Aerial Robotics

Tutorial 4 Flight Control 1

Control Part 1 Tutorial

Hover, linearisation, PD/PID and motor dynamics

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Question 1 : Hover and acceleration

A level UAV has mass m = 1.60 kg. Neglect drag.

(a) Find the thrust required for stationary hover.

(b) With this thrust and w = +0.30 m/s, does it stop?

(c) Find the thrust for upward acceleration 1.20 m/s².

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Solution 1 : Force balance and signs

$m\ddot{z} = mg - T$, so $T = m(g - \ddot{z})$.

(a) $T_{hover} = 1.60(9.81) = 15.696$ N ≈ 15.7 N.

(b) $\ddot{z} = 0$, but w stays +0.30 m/s: constant descent.

(c) $\ddot{z} = -1.20$ m/s².

$T = 1.60[9.81 - (-1.20)] = 17.616$ N ≈ 17.6 N.

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Question 2 : Linearise one rotor

$T = kT\,\Omega^2$; $kT = 2.00 \times 10^{-5}$ N/(rad/s)^2.

Operating speed $\Omega_0 = 400$ rad/s; increase $\delta\Omega = 20$ rad/s.

(a) Derive the first-order thrust-increment formula.

(b) Calculate approximate and exact thrust increases.

(c) Identify the neglected term and explain its effect.

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Solution 2 : Keep the first-order term

$T = kT(\Omega_0 + \delta\Omega)^2$

$T = kT\,\Omega_0^2 + 2kT\,\Omega_0\delta\Omega + kT(\delta\Omega)^2$

Subtract $T_0 = kT\,\Omega_0^2$:

$\delta T = 2kT\,\Omega_0\delta\Omega + kT(\delta\Omega)^2$

For small $\delta\Omega$: $\delta T \approx 2kT\,\Omega_0\delta\Omega$.

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Solution 2 : Numerical check

Approximate increase: $2(2 \times 10^{-5})(400)(20) = 0.320$ N.

Exact increase: $(2 \times 10^{-5})(420^2 - 400^2) = 0.328$ N.

Neglected term: $(2 \times 10^{-5})(20^2) = 0.00800$ N.

The linear estimate underestimates this increase.

$T_0 = 3.20$ N; exact final thrust = 3.528 N.

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Question 3 : Design a PD controller

Use $e = z^d - z$; $u_z = k_p e + k_d \dot{e}$; $T = m(g - u_z)$.

Assume constant reference and ideal thrust tracking.

(a) Derive the closed-loop error equation.

(b) Choose gains for $\omega_n = 3.00$ rad/s and $\zeta = 0.800$.

State the units of both gains.

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Solution 3 : Match the characteristic polynomial

For constant z^d: $\ddot{e} = -\ddot{z} = -u_z$.

Therefore $\ddot{e} + k_d\dot{e} + k_p e = 0$.

Match $s^2 + k_d s + k_p$ with $s^2 + 2\zeta\omega_n s + \omega_n^2$.

$k_p = 3^2 = 9.00\ s^{-2}$.

$k_d = 2(0.800)(3) = 4.80\ s^{-1}$.

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Question 3 continued : Command a climb

m = 1.60 kg; use the gains just calculated.

z^d = −2.00 m; z = −1.80 m; ż = +0.10 m/s.

(c) Calculate e, ė, u_z and commanded thrust.

(d) Explain the acceleration direction.

Does the UAV immediately start moving upwards?

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Solution 3 : Upward acceleration slows descent

$e = -2.00 - (-1.80) = -0.200$ m.

$\dot{e} = 0 - 0.10 = -0.100$ m/s.

$u_z = 9(-0.200) + 4.8(-0.100) = -2.28$ m/s².

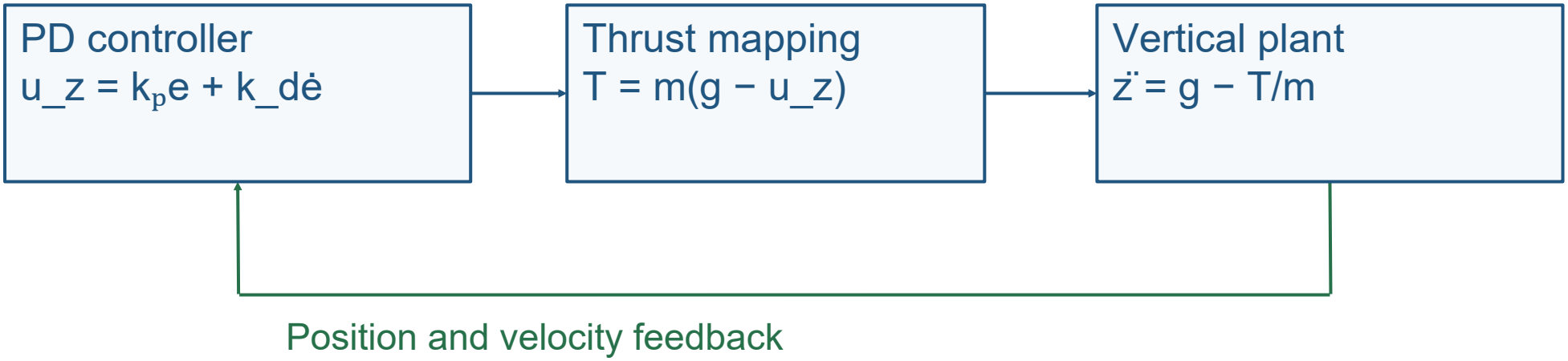
$T = 1.60(9.81 + 2.28) = 19.344$ N ≈ 19.3 N.

The UAV is still descending at this instant, but slowing down.

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Read the complete feedback path

Desired position and velocity enter the PD controller.



The controller asks for acceleration; the thrust mapping supplies the force.

The measured position and velocity determine the next command.

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Question 4 : Offset and integral action

Actual acceleration: $\ddot{z} = u_z + d_0$, with $d_0 = +0.45$ m/s².

Use $k_p = 9.00$ s⁻² and $k_d = 4.80$ s⁻¹.

(a) Find the PD steady-state error for constant reference.

(b) Explain how integral action can remove this error.

(c) Check $k_i = 10.0$ s⁻³ using $0 < k_i < k_d\, k_p$.

(d) Explain integral windup during actuator saturation.

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Solution 4 : Persistent error and compensation

$\ddot{e} + k_d\dot{e} + k_p e = -d_0$. At steady state: $e_{ss} = -d_0/k_p$.

$e_{ss} = -0.45/9 = -0.0500$ m: 5 cm below the target.

Integral action can supply $u_z = -d_0$ when $e = 0$.

For $k_i = 10$, the required integral state is $\eta = -0.0450$ m s.

This assumes a stable, unsaturated steady state.

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Solution 4 : Stability and windup

$k_d k_p = 4.80(9.00) = 43.2\ s^{-3}$.

$0 < 10.0 < 43.2$: the ideal no-lag PID model is stable.

Saturation prevents the actuator from delivering the command.

If error persists, the integral state may keep growing.

Conditional integration or back-calculation limits windup.

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Question 5 : Motor response and stability

First-order motor response: $\tau_m = 0.020$ s.

(a) Find the fraction of a speed step reached at 0.060 s.

For the local PD/lag model, $k_p = 100\ s^{-2}$; $k_d = 3.00\ s^{-1}$.

(b) Check stability using $k_d > \tau_m k_p$.

(c) Repeat for $\tau_m = 0.050$ s.

(d) Does stability guarantee satisfactory tracking?

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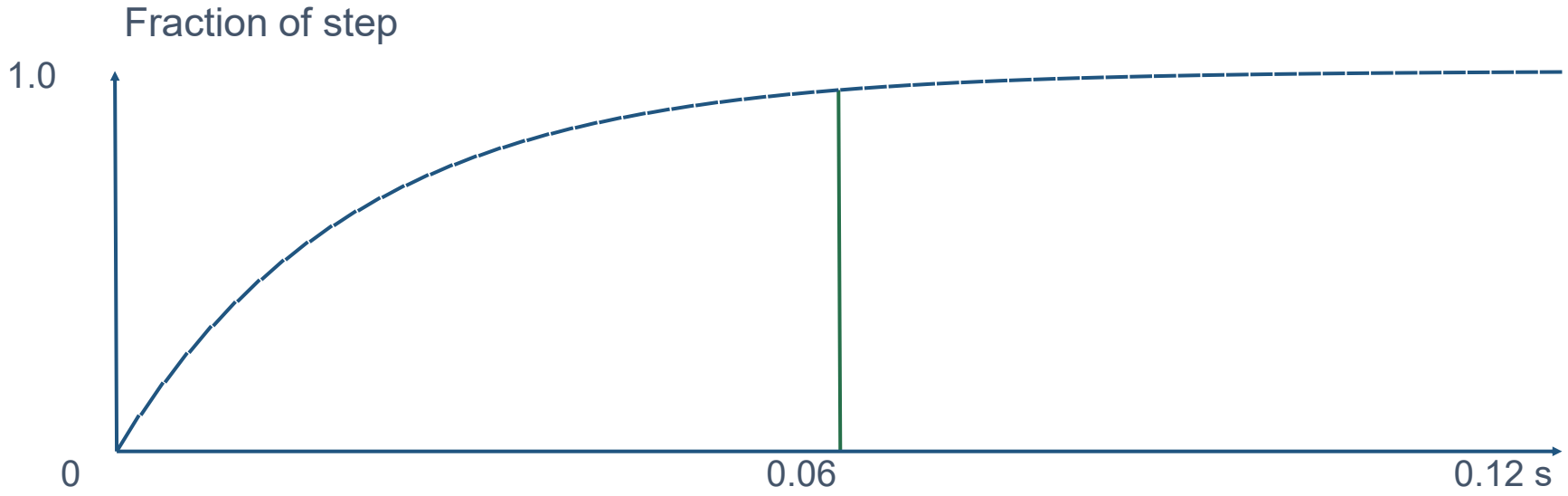
Solution 5 : Time response

Fraction reached = $1 - \exp(-t/\tau_m)$.

At t = 0.060 s: $t/\tau_m = 3$.

$1 - \exp(-3) = 0.9502 \approx 95.0\%$.

The response approaches its target gradually, not instantly.



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Solution 5 : Check the complete model

Characteristic polynomial: $\tau_m s^3 + s^2 + k_d s + k_p$.

For 0.020 s: $3.00 > 0.020(100) = 2.00 \rightarrow$ stable.

For 0.050 s: $3.00 < 0.050(100) = 5.00 \rightarrow$ unstable.

Stability does not specify overshoot, settling time or tracking error.

Also check actuator limits, noise and neglected dynamics.

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